

How to Install VarAC on Linux Mint (Wine 32)

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Introduction

This document describes the steps I followed to install **VARA HF** and **VarAC** on a **fresh installation of Linux Mint 22.3 Cinnamon**, using **Wine**.

Because VARA and VarAC behave differently depending on whether a **32-bit** or **64-bit Wine prefix** is used, I prepared **two separate guides**:

- *How to Install VarAC on Linux Mint (Wine 32-bit)* ⇐ *This document.*
- *How to Install VarAC on Linux Mint (Wine 64-bit)*

Each guide assumes:

- A clean Linux Mint installation
- A fresh Wine prefix (no leftovers from previous Wine setups)
- VARA HF v4.9.0 and VarAC v13.x

The steps documented here worked reliably on my system. They do not guarantee success on all environments, but they should help isolate common causes of startup crashes and configuration issues.

Environment

- Linux Mint 22.3 Cinnamon
- Wine 32-bit prefix under ~/.wine
- VARA HF v4.9.0
- VarAC V13.0.1

Installation Steps

Step 0: Preparation

1. Install Linux Mint 22.3.
2. After installation of Linux Mint, run below:
\$ sudo apt update
\$ sudo apt upgrade
\$ sudo apt autoremove
\$ sudo gpasswd -a <your_user> dialout
\$ sudo reboot

A reboot may not be necessary, but it ensures that you have been added to the dialout group.

Step 1: Install and setup Wine HQ (not the distro Wine)

1. Install wine-hq
Refer to <https://wine.htmlvalidator.com/install-wine-on-linux-mint-22.html>
But do not run winecfg here.

2. Create a 32-bit Wine prefix by running:
\$ export WINEARCH=win32
\$ wineboot -u

Install wine-mono, if prompted.

3. Run winecfg
\$ winecfg

Set the Windows version to "**Windows 10**".

Verify win32 by running:

```
$ file ~/.wine/drive_c/windows/system32/kernel32.dll  
/home/<your_user>/.wine/drive_c/windows/system32/kernel32.dll: PE32 executable  
(DLL) (console) Intel 80386, for MS Windows, 16 sections
```

4. Install winetricks:
\$ sudo apt install winetricks
\$ winetricks sound=alsa
\$ winetricks vb6run

The vb6run step is required for VARA.

Step 2: Replace pdh.dll with the Microsoft version

The Wine 32 version of pdh.dll does not work correctly with VARA.

1. Download the Microsoft version:
\$ wget http://download.microsoft.com/download/winntsrv40/update/5.0.2195.2668/nt4/en-us/nt4pdh.dll.exe
2. Execute the file with Wine, then copy the following files:
 - pdh.dll
 - pdh.dbg
 - pdh.pdbto:
~/.wine/drive_c/windows/system32

Step 3: Install and run VARA Modem

```
$ wine 'VARA setup (Run as Administrator).exe'
```

```
$ cd ~/.wine/drive_c/VARA  
$ wine VARA.exe
```

Step 4: Install Segoe UI font

This step is required for VarAC to display the user interface correctly.

```
$ git clone https://github.com/SpudGunMan/segoe-ui-linux  
$ cd segoe-ui-linux  
  
$ sudo mkdir -p /usr/share/fonts/Microsoft/TrueType/SegoeUI  
$ sudo cp font/*.ttf /usr/share/fonts/Microsoft/TrueType/SegoeUI/  
$ sudo fc-cache -fv  
  
$ cp font/*.ttf ~/.wine/drive_c/windows/Fonts/
```

Verify the installation:

```
$ fc-match "Segoe UI"
```

The expected output is:

```
segoeui.ttf: "Segoe UI" "Regular"
```

Note:

During testing on Linux Mint 22.3, the repository's `install.sh` reported a successful installation but did not correctly install the Segoe UI fonts. Therefore, the manual installation procedure above is recommended.

Step 5: Install and run VarAC

```
$ wine VarAC_Installer_V13_0_1.exe  
$ cd ~/.wine/drive_c/VarAC  
$ wine VarAC.exe
```

Check the com port by looking at ~/.wine/dosdevices.

If you see a symbolic link, for example:

```
com33 -> /dev/ttyUSB0
```

Com33 is the one you set at VarAC CAT control.

What matters is that it correctly points to the tty device of your radio - e.g., /dev/ttyUSB0, /dev/ttyACM0 and etc.

With the above steps, VarAC v13 starts and runs normally on Linux Mint 22.3 for me.

Now set the initial values as requested by VarAC.

Don't forget to enable "**Linux compatibility mode**" in:

Settings → RIG Control and VARA Configurations → Misc

Possible causes of the crash may include:

- 32-bit vs 64-bit Wine prefix
- Missing vb6run
- Missing Segoe UI font
- Leftover state from older Wine prefixes

Wine Tips

When you want purge all the current wine 32 environment and recreate it freshly, follow steps below:

- `rm -rf ~/.wine`
- `unset WINEPREFIX`
- `unset WINEARCH`
- `export WINEARCH=win32`
- `wineboot -u`

Please note that this removes all Wine applications and prefixes.

Hope this helps.

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